

IVAILO, Bulgaria

WMO: 156280

Lat: 42.217N

Lon: 24.333E

Elev: 702

StdP: 14.33

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	18.1	8.0	8.5	15.0	12.5	10.7	20.0	30.3	36.2	25.8	39.7	1.2	270	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	94.3	70.6	91.0	70.0	88.2	68.9	73.2	89.0	71.6	86.8	70.2	84.9	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	106.0	80.3	66.6	100.6	78.3	65.3	95.8	76.7	37.3	89.4	35.8	87.0	34.5	85.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.2	18.6	14.5	DB	4.9	100.5	7.3	4.2	-0.3	103.5	-4.5	106.0	-8.6	108.3	-13.9	111.4
			WB	4.4	76.8	7.3	2.7	-0.8	78.8	-5.1	80.4	-9.1	81.9	-14.4	83.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	34.0	37.8	45.4	53.5	63.2	70.4	74.7	74.2	65.9	55.7	44.9	35.4
	DBStd	15.87	8.25	8.47	7.44	6.10	5.50	5.34	4.41	4.38	5.36	6.98	7.26	7.99
	HDD50	1726	498	346	182	33	0	0	0	0	0	28	185	454
	HDD65	4661	961	761	608	345	100	15	1	2	52	296	603	917
	CDD50	3437	1	5	39	139	409	611	767	751	476	204	32	3
	CDD65	898	0	0	0	1	44	176	303	288	78	7	1	0
	CDH74	9668	0	0	5	46	577	1887	3264	2968	813	104	4	0
	CDH80	3942	0	0	0	7	129	730	1484	1329	249	14	0	0
(14) Wind	WSAvg	3.6	3.5	4.4	5.0	4.0	3.5	4.0	4.1	3.2	3.1	2.6	3.0	3.0
(15) Precipitation	PrecAvg	23.1	1.8	1.6	1.8	1.7	2.8	2.3	2.2	1.6	1.7	1.8	1.5	2.5
	PrecMax	32.3	3.6	3.9	4.8	3.4	6.5	5.0	6.1	5.3	3.7	4.4	4.4	5.2
	PrecMin	12.4	0.3	0.1	0.2	0.6	0.9	0.8	0.0	0.0	0.4	0.1	0.0	0.6
	PrecStd	4.9	0.8	1.0	1.1	0.9	1.5	1.1	1.5	1.3	0.9	1.1	1.1	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	65.4	72.9	79.7	87.7	94.9	99.5	98.4	90.9	82.0	71.5	60.9
		MCWB	48.9	51.7	56.2	62.2	67.4	71.0	70.6	71.3	68.0	64.8	58.2	52.6
	2%	DB	52.8	60.1	68.1	73.8	83.5	90.8	94.7	94.0	86.3	76.7	65.4	53.4
		MCWB	45.0	49.3	53.6	58.5	65.3	70.3	70.8	70.5	66.7	62.4	55.4	47.7
	5%	DB	48.5	55.4	63.6	70.0	80.3	87.4	91.1	90.6	82.5	72.3	60.0	49.3
		MCWB	43.1	47.3	51.2	56.4	63.8	68.9	70.4	69.8	65.4	60.0	52.4	45.8
	10%	DB	44.8	50.3	59.4	66.5	76.9	84.2	88.0	87.5	78.6	68.0	56.0	46.3
		MCWB	40.7	43.8	49.2	54.8	62.6	67.7	69.0	68.7	63.5	57.8	50.4	43.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	52.6	57.8	63.3	69.5	74.4	74.9	75.4	69.9	65.8	59.7	53.4
		MCDB	57.6	63.4	70.2	76.8	83.0	88.6	92.8	90.8	85.4	79.1	69.7	58.0
	2%	WB	46.0	49.7	54.3	60.0	67.3	72.3	73.0	73.2	68.1	63.3	56.3	48.9
		MCDB	50.9	59.2	66.0	71.5	80.4	86.7	89.3	89.0	82.8	74.4	63.0	51.8
	5%	WB	43.5	47.2	51.9	57.7	65.3	70.4	71.5	71.2	66.4	61.2	53.9	46.5
		MCDB	47.7	54.9	62.3	67.5	77.2	84.5	86.8	86.6	80.1	69.5	58.6	49.4
	10%	WB	41.2	44.3	49.5	55.7	63.5	68.8	70.1	69.6	64.7	59.2	51.0	43.7
		MCDB	44.6	49.3	58.5	64.7	74.8	81.8	85.1	84.2	75.8	65.6	55.3	46.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	16.0	19.5	20.8	22.4	23.4	24.6	24.9	22.3	18.9	16.0	12.3
		MCDBR	20.2	26.2	28.4	28.3	29.3	29.6	30.7	30.6	29.9	26.5	23.5	17.8
		MCWBR	13.6	16.9	16.1	15.2	13.7	12.0	11.5	11.7	13.6	14.0	14.1	12.8
		MCDBR	18.6	24.6	26.4	24.5	26.4	27.5	27.5	26.8	26.4	23.2	20.0	15.1
	5% WB	MCWBR	13.0	16.2	15.3	13.7	13.5	12.3	11.5	11.4	12.6	12.7	12.6	11.3
(40) Clear-Sky Solar Irradiance	taub		0.319	0.349	0.397	0.456	0.449	0.450	0.451	0.459	0.432	0.386	0.348	0.324
	taud		2.382	2.314	2.229	2.073	2.170	2.204	2.209	2.185	2.230	2.347	2.417	2.401
	Ebn at Noon		259	268	267	258	264	263	261	255	252	251	246	246
	Edh at Noon		26	32	40	50	46	45	44	44	39	31	25	23
(44) All-Sky Solar Radiation	RadAvg		532	789	1112	1435	1732	1914	2009	1828	1363	930	611	462
	RadStd		77	100	104	161	134	161	150	141	105	104	58	67

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+1.35	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	+0.83	N/A	N/A	N/A	N/A	+1.08	+1.10	-196	-281	N/A	N/A

Nomenclature: See separate page